

Monin–Obukhov Similarity Theory (MOST) & Stability Functions

Technical Reference Summary & Equation Cheat Sheet — Dr. David E. England

1. Fundamental Scales & Non-Dimensional Quantities

Parameter	Formula
Friction velocity	$u_* = \sqrt{\tau_0/\rho}$
Temperature scale	$\theta_* = -Q_0/u_* = -\overline{w'\theta'}_0/u_*$
Obukhov length	$L = \frac{u_*^2 \theta_0}{\kappa g \theta_*}$
Stability param.	$\zeta = z/L$ (< 0 unstable, $= 0$ neutral, > 0 stable)
Von Kármán const.	$\kappa \approx 0.40$
Excess resistance	$kB^{-1} \equiv \ln(z_{0,m}/z_{0,h}); \approx 1-2$ (smooth/water) to 10–12+ (arid/urban)

2. Point-Gradient Functions $\phi_{m,h}(\zeta)$

$$\frac{\kappa z}{u_*} \frac{\partial u}{\partial z} = \phi_m(\zeta), \quad \frac{\kappa z}{\theta_*} \frac{\partial \theta}{\partial z} = \phi_h(\zeta)$$

Unstable ($\zeta < 0$): $\phi_m = (1 - \beta_m \zeta)^{-1/4}$, $\phi_h = Pr_{t0}(1 - \beta_h \zeta)^{-1/2}$; $\beta_m = \beta_h = 16$, $Pr_{t0} = 1.0$

Stable ($\zeta \geq 0$): $\phi_m = 1 + \gamma_m \zeta$, $\phi_h = Pr_{t0} + \gamma_h \zeta$; $\gamma_m = \gamma_h = 5$, $Pr_{t0} = 1.0$

3. Integrated Profiles & Bulk Ri

$$u(z) = \frac{u_*}{\kappa} \left[\ln \frac{z}{z_{0,m}} - \Psi_m\left(\frac{z}{L}\right) + \Psi_m\left(\frac{z_{0,m}}{L}\right) \right]$$

$$\theta(z) - \theta_s = \frac{\theta_*}{\kappa} \left[\ln \frac{z}{z_{0,h}} - \Psi_h\left(\frac{z}{L}\right) + \Psi_h\left(\frac{z_{0,h}}{L}\right) \right]$$

$$Ri_g \equiv \frac{(g/\theta_0) \partial \theta / \partial z}{(\partial u / \partial z)^2} = \zeta \frac{\phi_h}{\phi_m^2}$$

$$Ri_b \equiv \frac{(g/\theta_0)[\theta(z_1) - \theta_s]z_1}{u(z_1)^2} = \frac{z_1}{L} \cdot \frac{\ln(z_1/z_{0,h}) - \Psi_h(z_1/L)}{[\ln(z_1/z_{0,m}) - \Psi_m(z_1/L)]^2}$$

4. Closed-Form Stability Functions $f_{m,h}(Ri)$

$$K_m = (\kappa z)^2 \left| \frac{\partial u}{\partial z} \right| f_m(Ri), \quad K_h = (\kappa z)^2 \left| \frac{\partial \theta}{\partial z} \right| f_h(Ri)$$

Mapping: $f_m = \frac{1}{\phi_m^2}$, $f_h = \frac{1}{\phi_m \phi_h}$, $\zeta = Ri \frac{f_h}{f_m^{3/2}}$
(England & McNider 1995 analytical branches)

• Stable ($Ri \geq 0$):

$$f_m = \left[\frac{1 - 2\gamma_m Ri + \sqrt{1 - 4(\gamma_m - \gamma_h) Ri}}{2(1 - \gamma_m Ri)} \right]^2, \quad Ri_c = \frac{1}{\gamma_m} = 0.20$$

• Unstable ($Ri < 0$): cubic in $x \equiv f_m^2$:

$$x^3 - 2x^2 + (1 - \beta_m \beta_h Ri^2)x - \beta_m(\beta_m - \beta_h) Ri^2 = 0$$

• Equal coeff. ($\beta_m = \beta_h = \beta$):

$$f_m(Ri) = \sqrt{1 + \sqrt{1 + \beta^2 Ri^2}}$$

• Near-neutral ($|Ri| \leq 0.01$):

$$f_m \approx 1 - \frac{1}{2} \beta_m Ri - \frac{1}{8} \beta_m (3\beta_m - 2\beta_h) Ri^2 \xrightarrow{\beta=16} 1 - 8Ri - 32Ri^2$$

5. Discrete Grid Depth Matching (z_{eff})

Equate Ri_b to $Ri_g(z_{\text{eff}})$ (implicit in z_{eff}):

$$z_{\text{eff}} = z_1 \left[\frac{\ln(z_1/z_{0,h}) - \Psi_h(z_1/L)}{(\ln(z_1/z_{0,m}) - \Psi_m(z_1/L))^2} \right] \frac{\phi_m^2(z_{\text{eff}}/L)}{\phi_h(z_{\text{eff}}/L)}$$

Neutral ($z_{0,m} \neq z_{0,h}$): $z_{\text{eff},n} = z_1 \frac{\ln(z_1/z_{0,h})}{[\ln(z_1/z_{0,m})]^2}$

Neutral ($z_{0,m} = z_{0,h} = z_0$): $z_{\text{eff},n} = \frac{z_1}{\ln(z_1/z_0)}$

6. Outer Boundary Layer Extensions

Blackadar mixing length ($z > 0.1z_i$):

$$\frac{1}{\ell_m(z)} = \frac{1}{\kappa z} + \frac{1}{\lambda_{0,m}} \implies \ell_m(z) = \frac{\kappa z}{1 + \kappa z/\lambda_{0,m}}, \quad \lambda_{0,m} \approx 0.00027 \frac{G}{f}$$

Free convection limit ($\zeta \rightarrow -\infty$, $u_* \rightarrow 0$):

$$w_* = \left(\frac{g}{\theta_0} Q_0 z_i \right)^{1/3}$$